INCH-POUND

MIL-DTL-22442/65
8 February 2024

DETAIL SPECIFICATION SHEET

CABLE ASSEMBLY, AIRCRAFT AUDIO

This specification sheet is approved for use by all
Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-22442.

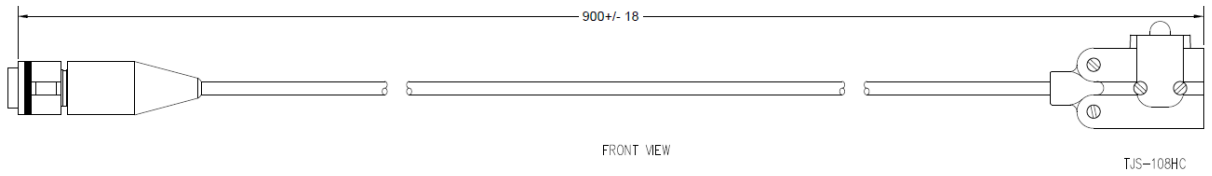


FIGURE 1. Cable assembly P/N M22442/XX-1.

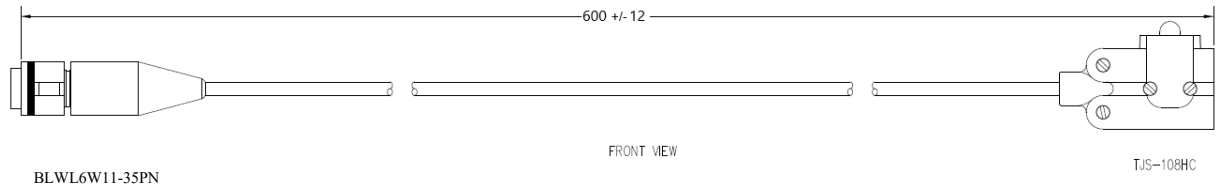


FIGURE 2. Cable assembly P/N M22442/XX-2.

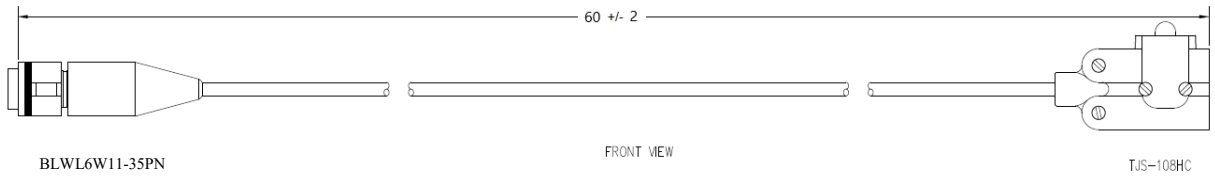


FIGURE 3. Cable assembly P/N M22442/XX-3.

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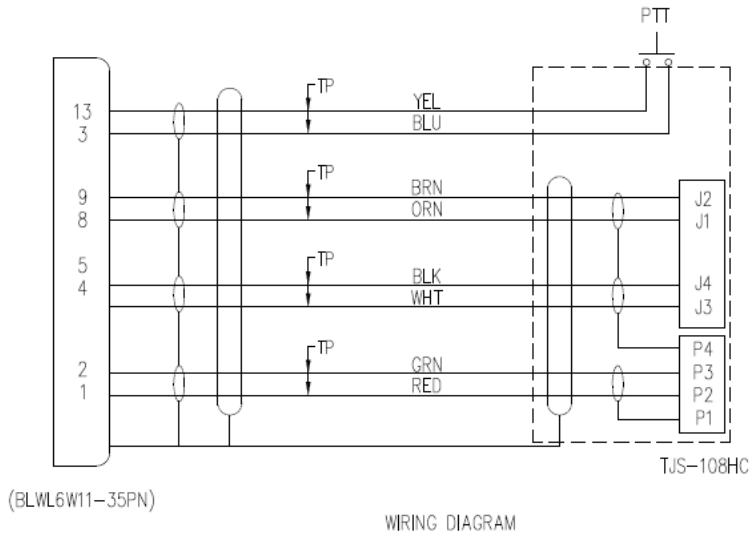


FIGURE 4. Cable assembly P/N M22442/XX-1, M22442/XX-2, and M22442/XX-3 schematic diagram.

NOTE TO FIGURES 1 THROUGH 4: Connector BLWL6W11-35 is available from G & H Technology, Inc., Aerospace Products Division, 750 West Ventura Blvd Camarillo, CA 93010, and connector TJS-108HC is available from Amphenol Nexus Technologies, 316 Cortland Ave., Stamford, CT 06906. Equivalent connectors to BLWL6W11-35 and TJS-108HC may be used if approved by the Qualifying Activity.

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REQUIREMENTS:

1. Conductors. Single conductors shall have a nominal cross-sectional area of 183 circular mils 19/40 (No. 28 AWG) hard strength cadmium free copper alloy. Conductors shall have a tin plating over the cadmium free copper alloy. The strands shall be twisted to have a left-hand lay of .50 inch nominal.

2. Color code. The conductor insulation for the M22442/XX-1, M2244/XX-2, and M22442/XX-3 cord assembly shall be colored red, green, black, white, brown, orange, yellow, and blue**.** The red-green, black-white, brown-orange, and yellow-blue shall be twisted, shielded pairs. Each twisted shielded pair shall have a thermoplastic rubber (TPR) isolator.

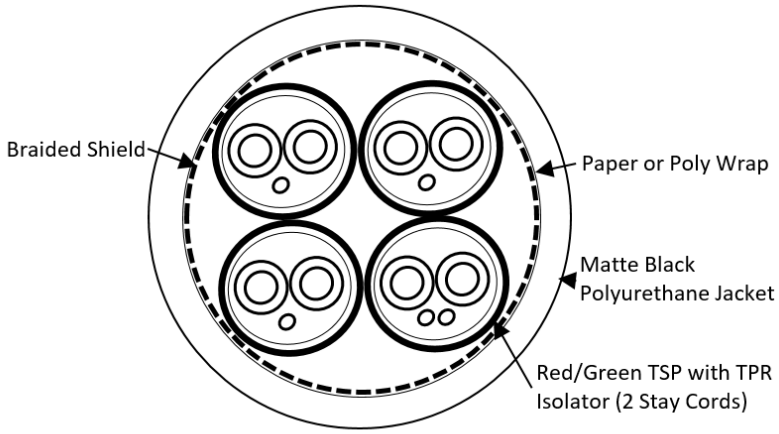


FIGURE 5. Cable construction P/N M22442/XX-1, P/N M2244/XX-2, and M22442/XX-3.

3. Stay cords. Each stay cord shall have a minimum breaking strength of 25 pounds and shall not abrade or otherwise damage the insulation when the cord is bent or stretched.

4. Jacketing materials elastomer. The M22442/XX-1, and -2 and -3 cable shall have a flame retarding polyurethane jacket. The jacket is to be applied in a continuous extrusion process or molded process directly onto the cable core. The jacket outside diameter (O.D.) shall be in the range of 0.280 to 0.320 inch diameter with a jacket wall thickness of 0.030 to 0.040 inch.

5. Shield. Each twisted shielded pair (TSP) shall be shielded with spiral wrapped shielding, 90-95 percent coverage, using size 40 AWG, tinned soft copper, wrapped with Mylar, and jacketed with nominal 0.007 inch thermoplastic rubber isolator. The cable shall be shielded overall with a braid of size 36 AWG tinned soft copper, min 90 percent coverage. There shall be a paper or poly wrap separator.

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6. Cable (cord) construction. Twisted shielded pairs shall be twisted together with a 0.5 inch nominal right-hand lay. Shielded, wrapped, and jacketed twisted shielded pairs shall be cabled with fillers as required. Twisted shielded pairs of cable assemblies M22442/XX-1, M22442/XX-2, and M22442/XX-3 shall be twisted together with a 1.25 inch nominal right-hand lay.

7. Cable conductor separation. The finished cable shall have all conductors within a single outer jacket.

8. Cable assembly diameter. The diameter of the cable assembly shall not be greater than 0.320 inches.

9. Stay cord. The stay cord shall be firmly attached to the connectors and/or strain relief.

10. DC resistance. The DC resistance shall be less than 2 ohms between the following contacts (Cable M22442/XX-1, M22442/XX-2, and M22442/XX-3):

(a) Contact 1 of P/N BLWL6W11-35PN (in accordance with MIL-DTL-38999/46) and contact P2 of TJS-108HC.
(b) Contact 2 of P/N BLWL6W11-35PN and contact P3 of TJS-108HC.
(c) Contact 4 of P/N BLWL6W11-35PN and contact J3 of TJS-108HC.
(d) Contact 5 of P/N BLWL6W11-35PN and contact J4 of the TJS-108HC.
(d) Contact 8 of P/N BLWL6W11-35PN and contact J1 of the TJS-108HC.
(d) Contact 9 of P/N BLWL6W11-35PN and contact J2 of the TJS-108HC.

11. TSP isolation. The red-green TSP electromagnetic isolation from all other TSPs shall be as specified in table I, measured in accordance with MIL-DTL-5898.

TABLE I. Isolation.

Frequency (Hz)	Electrostatic	Electromagnetic
	(dB) minimum	
7,000	140	90
5,000	140	95
1,000	140	100

12. Length. Cable assembly M22442/XX-1 shall have a length of 900 ± 18. M22442/XX-2 shall have a length of 600 ± 12. M22442/XX-3 shall have a length of 60 ± 2.

13. Flex life. The cord assembly shall have a mean flex life of 500,000 flexes without showing evidence of:

(a) Damage to the outside jacket.
(b) Damage to the insulation of the individual conductors.
(c) Continuity.

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VERIFICATION:

1. Cable assembly. The cable assembly shall be tested in accordance with the tests listed in table II.

2. Flex life. Five samples of the unterminated cord shall be clamped and suspended through holes in a metal bar having a rectangular cross section of .50 inch by .625 inch as described below:

(a) The clamping point shall be immediately above the metal bar; the cord clamp shall rest on top of the metal bar.
(b) The cord shall be clamped in such a manner that it will not turn in the hole during the flexing test.
(c) The length of the through holes in the metal bar shall be .625 inch. The diameters of the through holes shall be 0.005 to 0.010 inch greater than the diameter of the cord samples.
(d) The through holes in the metal bar shall incorporate a .125 inch radius at the end from which the cord emerges (bending point).
(e) A three-pound weight shall be hung from the cord at a point 8 to 12 inches below the bar to assure that the cord is held in a vertical position.
(f) A small low-current drain, a 6-volt lamp, and a suitable power source shall be connected in series to each connected in series to each conductor to indicate electrical continuity (see figure 6). The metal bar, supported horizontally, shall then be rotated about its longitudinal axis back and forth through an angle of 120 degrees (60 degrees each side of vertical) at a rate of approximately 85 cycles per minute. The arithmetical mean of the flexing cycles of the five samples of the cord before electrical discontinuity occurs shall be taken as the flexing life of the cord (see 3.5.1 of MIL-DTL-22442).

3. Pull Test, Connectors. A pull force of 40 ± 2 pounds shall be applied.

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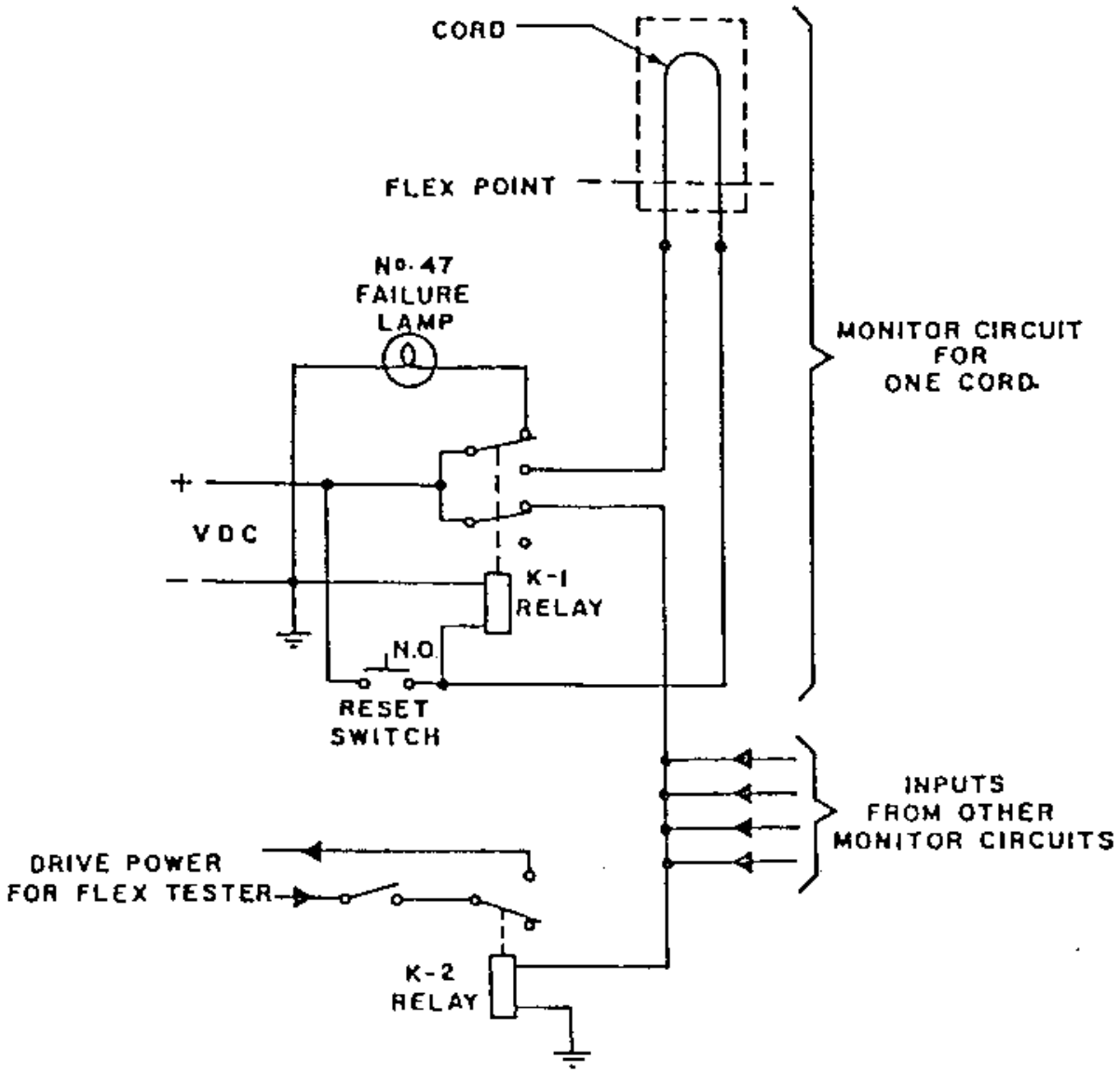


FIGURE 6. Monitor circuit for cord flex tester.

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TABLE II. Tests.

Inspection	Requirement Paragraph of MIL-DTL-22442	Test Paragraph	Group I Qualification Tests	Group II Quality Conformance Test 100%	Group III Sample Testing Per 4.4.1
GROUP A (ALL SAMPLE UNITS) 1/					
Visual & Mechanical Inspection	3.2 & 3.4	4.5.2	X	X	
Soldering	3.4.18	4.5.2.1	X	X	
Continuity	3.5.1	4.5.3	X	X	
GROUP B					
Dielectric Withstanding Voltage	3.5.2	4.5.4	X		X
Gauging 1/	3.5.3	4.5.5	X		X
Flex Life	2/	2/	X		X
Pull and Twist Test	3.5.4	4.5.6	X		X
GROUP C					
Low Temperature Flexibility	3.5.6	4.5.8	X		X
Durability 1/	3.5.7	4.5.9	X		X
GROUP D 1/					
Salt Fog	3.6.2	4.6.2	X		
Vibration	3.6.4	4.6.4	X		
Humidity	3.6.5	4.6.5	X		
Thermal Shock	3.6.5	4.6.6	X		

1/ Minimum tests for QPL retention (see 4.3 of MIL-DTL-22442)
2/ The flex life requirement is requirement 13, and the verification requirement is verification requirement 2, both of this specification sheet.

NOTES
1. All dimensions are in inches.
2. Tolerances are as follows:
   Metal parts ± .005
   Molded sections ± .05.

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Referenced documents: In addition to MIL-DTL-22442, this document references the following:

MIL-DTL-5898
MIL-DTL-38999/46

CONCLUDING MATERIAL

Custodian:
Navy – AS
Air Force - 85

Preparing activity:
Navy - AS
(Project 5995-2024-001)

NOTE: The activity listed above was interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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